Analyzing Movie Reviews Across Genres and Reviewer Types

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Contributions:

Evan - Scraped and Processed all IMDb Data, Fit VADER Model

Elpidio - Scraped Professional Reviews, Created all Visualizations

Roy - Scraped Professional Review Ratings, Fit Roberta Model

(All of us worked together to write our respective sections of the report.)

Abstract:

This paper investigates the differences between professional and general user reviews among several genres of movies. We obtained movie review data from www.imdb.com and rogerebert.com and then conducted sentiment analysis on the reviews to find any variation in sentiment between genres and reviewers. We ultimately found that general reviewers commented more on their overall enjoyment of each film while professional reviewers were much more critical on each film. The professionals focused on smaller parts of each film evaluating the filmmaking quality and storyline behind each film.

Introduction

In the film industry, movie reviews significantly shape public perception and influence box office performance. Reviews from both seasoned critics and casual viewers offer deep insights into a film's impact on the public. This study utilizes data from IMDb and RogerEbert.com to analyze sentiments across different platforms. We aim to turn online opinions into quantifiable data, examining how genres are perceived and the differences between professional critics and general users. Our project addresses three key questions:

- How do sentiments in movie reviews vary across genres?
- How do reviews differ between professional critics and general users?
- Is there a correlation between critic ratings and audience ratings?

To conduct this analysis, the study will utilize data pulled from IMDb and RogerEbert.com. IMDb is a premier site for movie information that provides extensive user

reviews and ratings. RogerEbert.com, named after the renowned film critic, is known for its professional critic reviews and deep analytical content, offering detailed commentary on film quality and viewer reception. These platforms have been selected for their credibility and wide selection of movie reviews, providing a reliable base for analyzing a wide range of opinions and sentiments across the movie reviewing landscape.

Data Acquisition and Processing

Scraping Methodology & Processing:

IMDb:

When we first started our project we looked to scrape IMDb but kept getting 403 errors which would not allow us to get the data. We added a User-Agent header which allowed us to finally begin to access the data on the IMDb page. After parsing the result using lxml, we found similar patterns for each 'href' link on the page and wrote an XPath selector that could pull these individual href links. After exploring the "By Genre" pages we noticed each genre had the same "Popular Movies" section which included trending movies for that genre. Even though the popular movies were a part of a sliding bar, all the links were available on the same html page and we did not need to use anything other than requests to select those links. Eventually we created a "get movies" function that could pull all the trending movies just by being given a genre link.

After getting the movie links, our next step was to find a way to get all the reviews for each movie link which proved to be much more challenging. In our first attempt to get the

reviews, we discovered the full form of each review was available on a separate page which could be accessed by each link on the reviews section. We pulled all the links available on the page using a common XPath for the review links and stored them in a list. Then we took this list of links and attempted to iterate through it, pulling the text and score of each review to store in sets. Unfortunately, many of the reviews were marked by what IMDb calls “spoiler warnings”. When a review on IMDb is marked as a spoiler warning the page will not show the review right away but only the first sentence or so. The rest of the review is hidden behind an icon which you can click to extend the page. This proved to be a large obstacle because the text behind the spoiler icon was not accessible through our get request. We looked through all the html in the sections we expected the text to be and came up empty handed. We determined we needed more versatility than requests and lxml could offer and decided to try using a selenium webdriver to click the expand text icon and access the full reviews. This ended up being our final strategy to get all the reviews. We identified the general class of the expander icon and used selenium’s “find_element” and “.click()” methods to first find the icon using its XPath, then click it to load the rest of the review. We noticed it took a bit after clicking for the driver to actually load the review data so we used a time.sleep function from the time package to give the driver time to load the page. Then we again used the find_elements method to get the full review and also the user’s IMDb rating for the movie. We took the review result and concatenated each paragraph into one continuous string. This worked perfectly for reviews with the spoiler warnings but was throwing up errors whenever there was no button for the program to find. This resulted in us adding try/except statements in case there was no button to click. We also faced issues as some people did not leave an IMDb rating on their review so we put in another try/except statement in

order to account for no rating being present. After figuring out that this method worked on one genre, we put it all into a function so it could be used to get reviews for any genre.

After figuring out how we could get all the reviews for each genre, we moved on to start processing each review as they had many new line characters, <p> elements, and unnecessary stop words which would be necessary to remove in order to conduct sentiment analysis on each review later on. To do this we wrote a function that would process each review by converting it to lower case then removing newlines and special characters using the re package. Finally we used nltk to get a list of common English stop words and remove them from the cleaned reviews.

To bring all of these together, we wrote a function, get_movie_data(), that used a request to obtain the title and average IMDb rating of any movie. We got both of these using their XPath from the movie page and stored them in variables. Then we called the get_reviews() function to get all the reviews for that movie and the process_reviews() function to process the reviews so we could work with them later on. This function returns the title of the movie, the average IMDb rating, every review for that movie and the score of each individual review.

After putting all these functions together we would paste the link of a genre into the get_movies() function to get all popular movies for that genre. After getting the movies, we just wrote a for loop that would iterate through each movie and collect the data needed using the get movie data on each movie. Then we wrote a function to store all of these movies into one pandas DataFrame and saved each DataFrame as a csv labeled by genre to use later.

RogerEbert:

Once we properly scraped and retrieved user reviews from IMDb, the next step for our group was to look to compare these general user reviews to a more professional source, and so we scraped reviews from Roger Ebert's website. We first created a get_professional_reviews()

function that used regex to properly extract only movie titles from the website, meaning we had to remove any unnecessary information such as release year, or “IMDb” from these movie titles. After extracting just the movie title itself, we sent a search request to the website’s undocumented api which was in the form of a search bar to look for professional reviews that matched the newly cleaned movie title. If there was a match, the website would then return the proper link that took us to the “Movie Reviews” section of the website. From there, we parsed the response using XPath and specifically looked for <a> tags on the website so that we could retrieve the first review link after putting in the movie title.

The next step was to do the actual text scraping of the review as well as properly scraping the rating for the movie. First, the actual scraping of the text from the review. We had to scrape all of the paragraphs and text of the review which were the <p> marks that were inside the <article> section of the website, and then turn that into one single long continuous string at the end with .join(review_text). We had an issue at one point where if we scraped the reviews from one genre the function would only return empty lists when we tried to scrape reviews for a different genre, this led us to initializing a session which fixed our problem and the function was back to working like normal.

Moving forward, the next step was to scrape the ratings based on the Roger Ebert website as well, and this is where we ran into challenges in terms of the comparison between IMDb and Roger Ebert. As previously stated, the IMDb ratings are on a scale out of 10 stars as opposed to the Roger Erbert rating system which is on a scale out of 4 stars. We also noticed that there was no actual number for how many stars each review received but only an image that was filled with 0.5, 1, 1.5…, up to 4 stars. Upon further inspection of the HTML, we noticed the class of the image containing the filled stars changed based on how many stars the movie received. This was

a lifesaver as we were able to write an XPath that selected the image and retrieved its class, then process it using regex to only get the score from each class labeled in the pattern, “h-7 filled star##”. We noticed each class ended with a rating out of 40 with 5 = 0.5 stars and 40 = 4 stars. This was a different rating scale than that of IMDb, so to properly compare the ratings from IMDb and Roger Ebert, we decided to convert the Roger Erbert ratings to a system out of 10 stars so that Roger Erbert matched IMDb. To do so, we first extracted the score out of 40 from the website, and turned it into an integer. Then, we took that rating and divided it by 40, and then multiplied it by 10 to get our finalized rating that was proportional to those from the IMDb website. But after properly adjusting the rating scales, at the end, the function can take any movie and return the cleaned movie title as well as the rating out of 10 and the full review from the professional movie viewer.

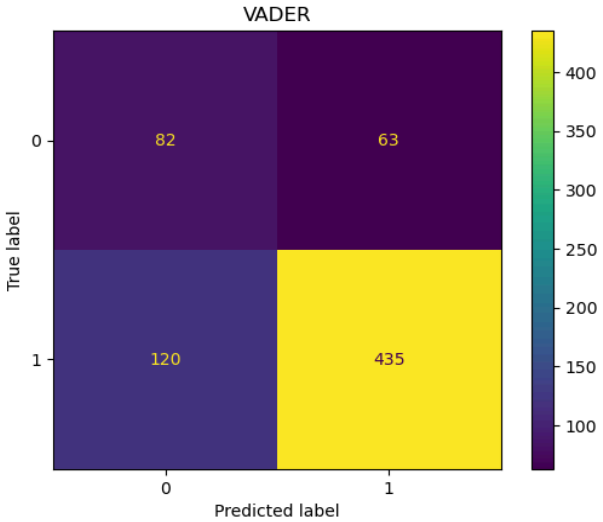
Putting everything together, our next step was being able to scrape the ratings and reviews from an entire dataset of movies, rather than just one movie as from the get_professional_reviews() function. This is when we made our rogerebert_review_machine() function that loops through the professional reviews of movies from the Roger Ebert website given that the movies were put into each data frame.

Visualizations, Methodology, and Descriptive Statistics

Our project uses sentiment analysis and other techniques to explore the sentiments and ratings of movie reviews across various genres, using VADER and Roberta Model NLP models, as well as other additional data visualizations.

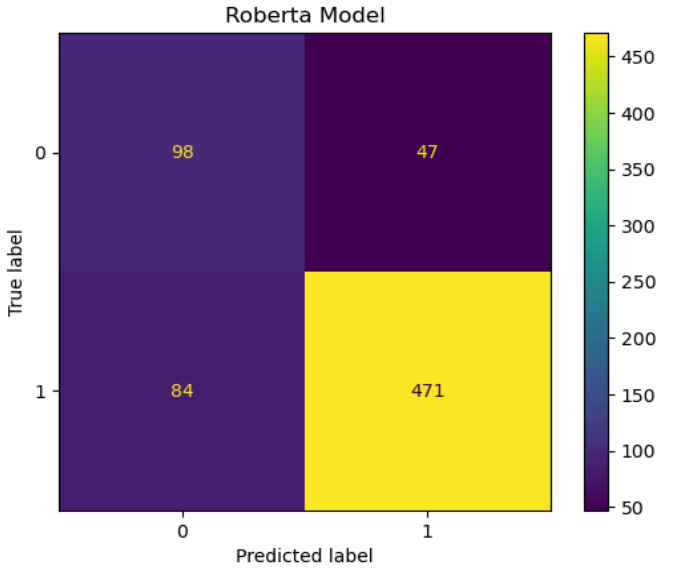
VADER Sentiment Analysis:

We conducted sentiment analysis on IMDb and Roger Ebert reviews using the VADER model, similar to the one discussed during lecture. This model uses a bag of words technique to classify movie reviews as either positive or negative based on text sentiment, giving a score of 1 for positive and 0 for negative reviews. We defined reviews with ratings of 6 stars or higher as positive and reviews with 5 stars or lower as negative. After converting these ratings into binary form, we used a confusion matrix to compare our VADER model's predictions with the actual ratings. The results showed a True Positive Rate of 0.784 and a False Positive Rate of 0.4344, indicating that while there were some issues with predicting negative reviews, the VADER model was generally effective in accurately classifying the sentiments of the movie reviews.



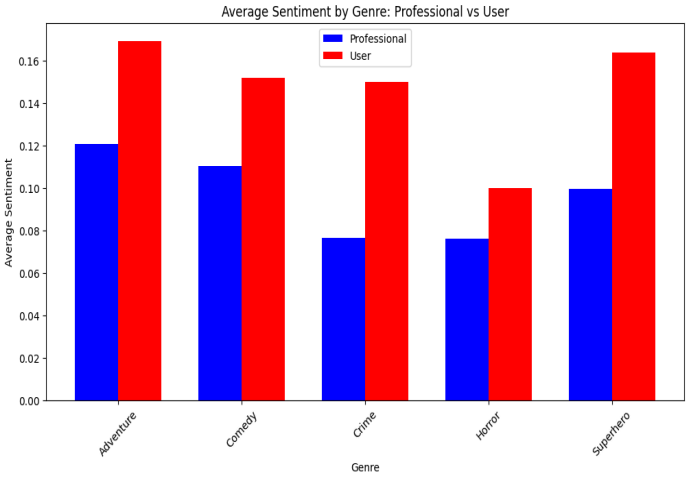
Roberta Model NLP:

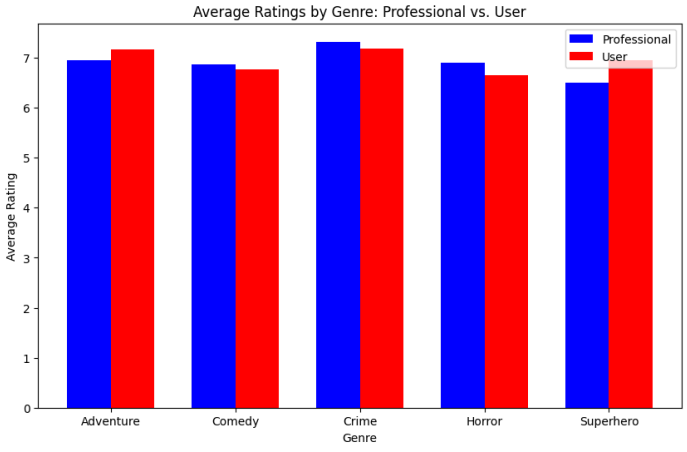
Another sentiment analysis we decided to use was a pre-trained Roberta Model NLP from Hugging Face. Due to its 512 token limit, we first had to make a split_text() function to break down the reviews into chunks and tokens. This was to make sure that no text from the reviews was excluded. We then analyzed these chunks with the Roberta Model, giving 1 to both positive and neutral reviews, and 0 to negative reviews. This model used the majority rule, where the majority of the sentiment in the tokens



decided the overall review as positive or negative. We applied the same criteria to classify the IMDb and Roger Ebert movie ratings, with scores of 6 stars and above being positive, and 5 stars and below as negative. The results showed the Roberta Model achieved a True Positive Rate of 0.849 and a False Negative Rate of 0.324, indicating it was generally more accurate than the VADER model, particularly in identifying negative sentiments with fewer errors in false negatives and positives, and a higher success rate in true positives and negatives.

Data Visualization





Our analysis includes histograms (appendix) that illustrate the distribution of sentiment scores across genres for both professionals and users. These histograms show that users generally exhibit a wider range of sentiment scores, suggesting a diverse reception compared to the narrower distributions seen with professionals. In addition, our scatter plots and correlation matrices (appendix) for each genre show different degrees of correlation between professional and user ratings. The correlation coefficients range from moderate to low across genres, indicating that while there is some alignment, users often rate movies differently compared to professional critics. Our bar charts further support this by comparing average ratings and sentiments across genres between professional critics and users,

revealing that while users overall exhibit higher sentiment scores, their ratings do not consistently surpass those of professionals. This is the case when looking at the average ratings of the comedy, crime, and horror genre.

Descriptive Statistics

Our statistics across all genres reveals mixed results in how users and professionals rate films. While in the Superhero and Adventure genre, users have a higher mean score compared to professionals, this trend does not consistently hold across all genres. For example, in Comedy, Crime, and Horror, professional ratings have higher mean scores than user ratings. Variability among user ratings is generally lower than among professionals across all genres, as indicated by smaller standard deviations. This suggests that user ratings tend to cluster more tightly around the mean, indicating less divergence in opinion compared to professional critics.

Descriptive Statistics:

	Adventure		Comedy		Crime		Horror		Superhero	
	Professional	User	Professional	User	Professional	User	Professional	User	Professional	User
count	30.000000	710.000000	28.000000	708.000000	28.000000	668.000000	29.000000	700.000000	29.000000	702.000000
mean	6.958333	7.161268	6.875000	6.762712	7.321429	7.177994	6.896552	6.643286	6.508621	6.946724
std	2.194314	1.033422	2.110928	0.813009	2.446518	1.261200	1.588909	0.918743	2.200327	1.307777
min	2.500000	4.700000	2.500000	4.700000	1.250000	4.600000	3.750000	4.400000	2.500000	4.000000
25%	5.312500	6.600000	6.250000	6.300000	5.937500	6.300000	5.000000	6.300000	5.000000	6.000000
50%	7.500000	7.000000	7.500000	6.900000	7.500000	7.100000	7.500000	6.600000	6.250000	6.900000
75%	8.750000	7.900000	8.750000	7.200000	8.750000	8.200000	7.500000	7.100000	7.500000	8.000000
max	10.000000	8.900000	10.000000	8.200000	10.000000	9.200000	10.000000	8.600000	10.000000	9.000000

Insights and Analysis

- **Variation of Sentiments Across Genres:** Our findings indicate that genres such as Adventure and Superhero consistently received more positive responses from users,

fitting with their die-hard fans. Conversely, genres like Horror showed more varied sentiments and ratings, showing their fans have more diverse opinions.

- **Differences Between Professional Critics and General Users:** There is a consistent difference in ratings and sentiments across genres, suggesting that professional reviewers use stricter criterias when making reviews, focusing on things like narrative depth and film execution. This is supported by the sentiment analysis from both the VADER and Roberta models.
- **Correlation Between Critic and Audience Ratings**:Our correlation analysis shows mixed results, with some genres aligning well between critic and user ratings, while others differ greatly. These findings indicate that perceptions of film quality differ across genres, shaped by the sentiment analysis our models offer. (Shown in Appendix).

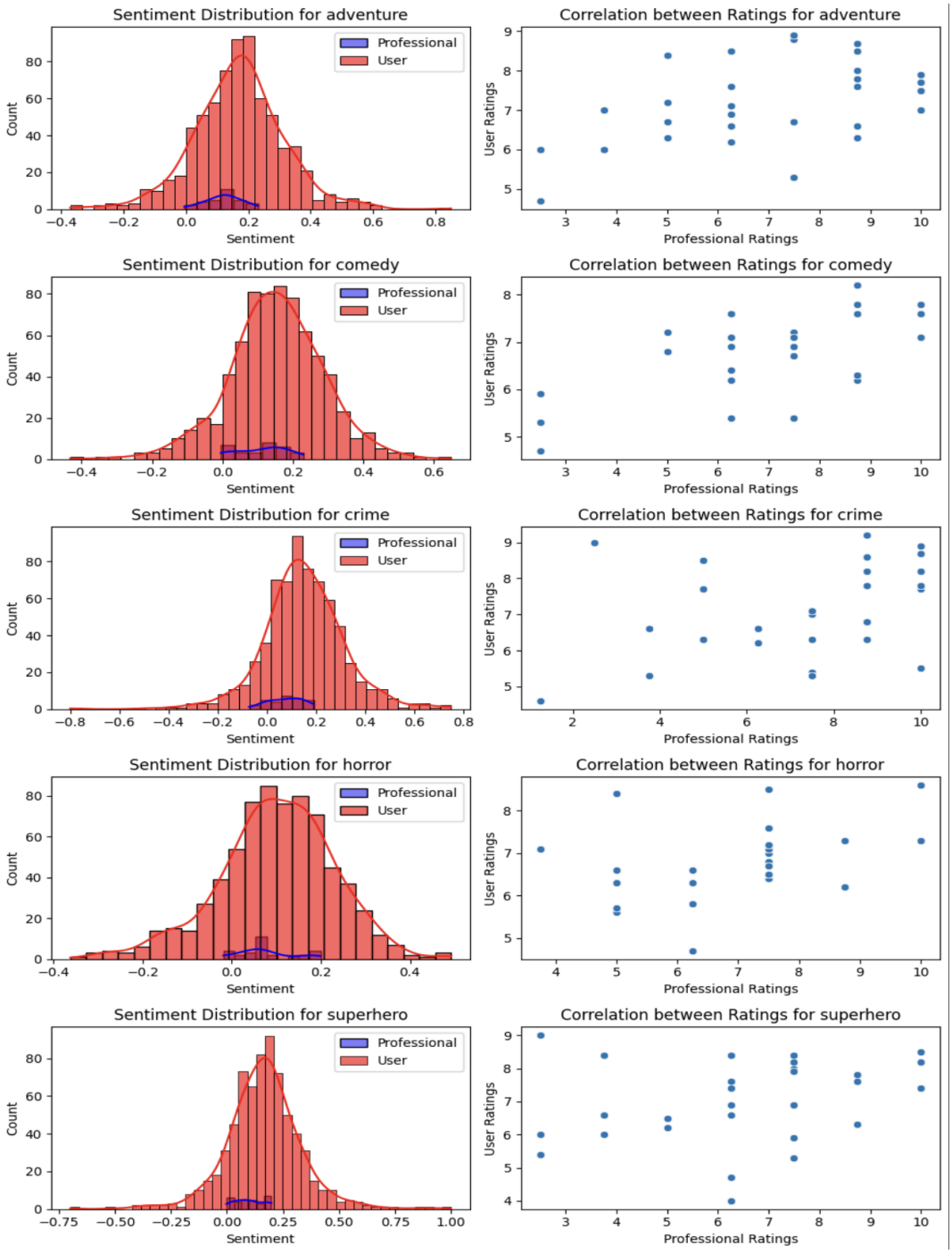
Conclusion:

Our analysis focused on the sentiment and rating dynamics within movie reviews across the selected popular genres of Adventure, Comedy, Crime, Horror, and Superhero. We successfully scraped data from IMDb and RogerEbert.com and then matched their rating scales. After that, we applied a VADER and Roberta model for sentiment analysis. The Roberta Model showed better accuracy when compared to the VADER model in terms of the overall predictions of positive and negative reviews. We also ran various forms of EDA, such as correlation studies and trend investigations, which supported our findings and helped us better understand the dynamics between general and professional viewers. The differences that we found showed that professional reviewers tended to be more critical, focusing on factors such as narrative and filmmaking quality, whereas general reviewers focused more on overall enjoyment.

Limitations:

The final result of our project gave us accurate results in terms of reviews and ratings from movies and analyzing them, however we still faced a range of limitations. For example, scraping all of the reviews and ratings from our target movie genres took an excessive amount of time to run the code. Each genre of movie that we decided to scrape required at least 40 minutes to run the code. With this time limitation, we were not able to scrape from all movie genres, so instead we focused on the more popular movie genres among general and professional viewers. Had we found an even more efficient way to scrape movies, we would have been able to pull a much more variety of information and compare general and professional reviews to a wider range of movies and genres.

Appendix:


Sentiment Distribution for adventure
Professional
User
Count
Sentiment
Correlation between Ratings for adventure
User Ratings
Professional Ratings
Sentiment Distribution for comedy
Professional
User
Count
Sentiment
Correlation between Ratings for comedy
User Ratings
Professional Ratings
Sentiment Distribution for crime
Professional
User
Count
Sentiment
Correlation between Ratings for crime
User Ratings
Professional Ratings
Sentiment Distribution for horror
Professional
User
Count
Sentiment
Correlation between Ratings for horror
User Ratings
Professional Ratings
Sentiment Distribution for superhero
Professional
User
Count
Sentiment
Correlation between Ratings for superhero
User Ratings
Professional Ratings

```
Correlation matrix for adventure Genre:
             Pro Rating   Rating
Pro Rating     1.00000  0.47073
Rating         0.47073  1.00000
Correlation matrix for comedy Genre:
             Pro Rating    Rating
Pro Rating    1.000000  0.630566
Rating        0.630566  1.000000
Correlation matrix for crime Genre:
             Pro Rating    Rating
Pro Rating    1.000000  0.406959
Rating        0.406959  1.000000
Correlation matrix for horror Genre:
             Pro Rating    Rating
Pro Rating    1.000000  0.345965
Rating        0.345965  1.000000
Correlation matrix for superhero Genre:
             Pro Rating    Rating
Pro Rating    1.000000  0.254592
Rating        0.254592  1.000000
```